

Call Buying

The success of a call buying strategy depends primarily on one's ability to select stocks that will go up and to time the selection reasonably well. Thus, call buying is not a strategy in the same sense of the word as most of the other strategies discussed in this text. Most other strategies are designed to remove some of the exactness of stock picking, allowing one to be neutral or at least to have some room for error and still make a profit. Techniques of call buying are important, though, because it is necessary to understand the long side of calls in order to understand more complex strategies correctly.

Call buying is the simplest form of option investment, and therefore is the most frequently used option "strategy" by the public investor. The following section outlines the basic facts that one needs to know to implement an intelligent call buying program.

WHY BUY?

The main attraction in buying calls is that they provide the speculator with a great deal of leverage. One could potentially realize large percentage profits from only a modest rise in price by the underlying stock. Moreover, even though they may be large percentage-wise, the risks cannot exceed a fixed dollar amount—the price originally paid for the call. Calls must be paid for in full; they have no margin value and do not constitute equity for margin purposes. Note: The preceding statements regarding payment for an option in full do not necessarily apply to very long-term (LEAPS) options, which were declared marginable in 1999. The following simple example illustrates how a call purchase might work.

Example: Assume that XYZ is at 48 and the 6-month call, the July 50, is selling for 3. Thus, with an investment of \$300, the call buyer may participate, for 6 months, in a move upward in the price of XYZ common. If XYZ should rise in price by 10 points (just over 20%), the July

50 call will be worth at least \$800 and the call buyer would have a 167% profit on a move in the stock of just over 20%. This is the leverage that attracts speculators to call buying. At expiration, if XYZ is below 50, the buyer's loss is total, but is limited to his initial \$300 investment, even if XYZ declines in price substantially. Although this risk is equal to 100% of his initial investment, it is still small dollarwise. *One should normally not invest more than 15% of his risk capital in call buying*, because of the relatively large percentage risks involved.

Some investors participate in call buying on a limited basis to add some upside potential to their portfolios while keeping the risk to a fixed amount. For example, if an investor normally only purchased low-volatility, conservative stocks because he wanted to limit his downside risk, he might consider putting a small percentage of his cash into calls on more volatile stocks. In this manner, he could "trade" higher-risk stocks than he might normally do. If these volatile stocks increase in price, the investor will profit handsomely. However, if they decline substantially—as well they might, being volatile—the investor has limited his dollar risk by owning the calls rather than the stock.

Another reason some investors buy calls is to be able to buy stock at a reasonable price without missing a market.

Example: With XYZ at 75, this investor might buy a call on XYZ at 80. He would like to own XYZ at 80 if it can prove itself capable of rallying and be in-the-money at expiration. He would exercise the call in that case. On the other hand, if XYZ declines in price instead, he has not tied up money in the stock and can lose only an amount equal to the call premium that he paid, an amount that is generally much less than the price of the stock itself.

Another approach to call buying is sometimes utilized, also by an investor who does not want to "miss the market." Suppose an investor knows that, in the near future, he will have an amount of money large enough to purchase a particular stock; perhaps he is closing the sale of his house or a certificate of deposit is maturing. However, he would like to buy the stock now, for he feels a rally is imminent. He might buy calls at the present time if he had a small amount of cash available. The call purchases would require an investment much smaller than the stock purchase. Then, when he receives the cash that he knew was forthcoming, he could exercise the calls and buy the stock. In this way, he might have participated in a rally by the stock before he actually had the money available to pay for the stock in full.

RISK AND REWARD FOR THE CALL BUYER

The most important fact for the call buyer to realize is that he will normally win only if the stock rises in price. All the worthwhile analysis in the world spent in selecting which call to buy will not produce profits if the underlying stock declines. However, this fact

should not dissuade one from making reasonable analyses in his call buying selections. Too often, the call buyer feels that a stock will move up, and is correct in that part of his projection, but still loses money on his call purchase because he failed to analyze the risk and rewards involved with the various calls available for purchase at the time. He bought the wrong call on the right stock.

Since the best ally that the call buyer has is upward movement in the underlying stock, the selection of the underlying stock is the most important choice the call buyer has to make. Since timing is so important when buying calls, the technical factors of stock selection probably outweigh the fundamentals; even if positive fundamentals do exist, one does not know how long it will take in order for them to be reflected in the price of the stock. One must be bullish on the underlying stock in order to consider buying calls on that stock. Once the stock selection has been made, only then can the call buyer begin to consider other factors, such as which striking price to use and which expiration to buy. The call buyer may have another ally, but not one that he can normally predict: If the stock on which he owns a call becomes more volatile, the call's price will rise to reflect that change.

The purchase of an out-of-the-money call generally offers both larger potential risk and larger potential reward than does the purchase of an in-the-money call. Many call buyers tend to select the out-of-the-money call merely because it is cheaper in price. *Absolute dollar price should in no way be a deciding factor for the call buyer.* If one's funds are so limited that he can only afford to buy the cheapest calls, he should not be speculating in this strategy. If the underlying stock increases in price substantially, the out-of-the-money call will naturally provide the largest rewards. However, if the stock advances only moderately in price, the in-the-money call may actually perform better.

Example: XYZ is at 65 and the July 60 sells for 7 while the July 70 sells for 3. If the stock moves up to 68 relatively slowly, the buyer of the July 70—the out-of-the-money call—may actually experience a loss, even if the call has not yet expired. However, the holder of the in-the-money July 60 will definitely have a profit because the call will sell for at least 8 points, its intrinsic value. The point is that, percentage-wise, *an in-the-money call will offer better rewards for a modest stock gain, and an out-of-the-money call is better for larger stock gains.*

When risk is considered, the in-the-money call clearly has less probability of risk. In the prior example, the in-the-money call buyer would not lose his entire investment unless XYZ fell by at least 5 points. However, the buyer of the out-of-the-money July 70 would lose all of his investment unless the stock advanced by more than 5 points by expiration. Obviously, the probability that the in-the-money call will expire worthless is much smaller than that for the out-of-the-money call.

The time remaining to expiration is also relevant to the call buyer. If the stock is fairly close to the striking price, the near-term call will most closely follow the price movement

of the underlying stock, so it has the greatest rewards and also the greatest risks. The far-term call, because it has a large amount of time remaining, offers the least risk and least percentage reward. *The intermediate-term call offers a moderate amount of each, and is therefore often the most attractive one to buy.* Many times an investor will buy the longer-term call because it only costs a point or a point and a half more than the intermediate-term call. He feels that the extra price is a bargain to pay for three extra months of time. This line of thought may prove somewhat misleading, however, because most call buyers don't hold calls for more than 60 or 90 days. Thus, even though it looks attractive to pay the extra point for the long-term call, it may prove to be an unnecessary expense if, as is usually the case, one will be selling the call in two or three months.

CERTAINTY OF TIMING

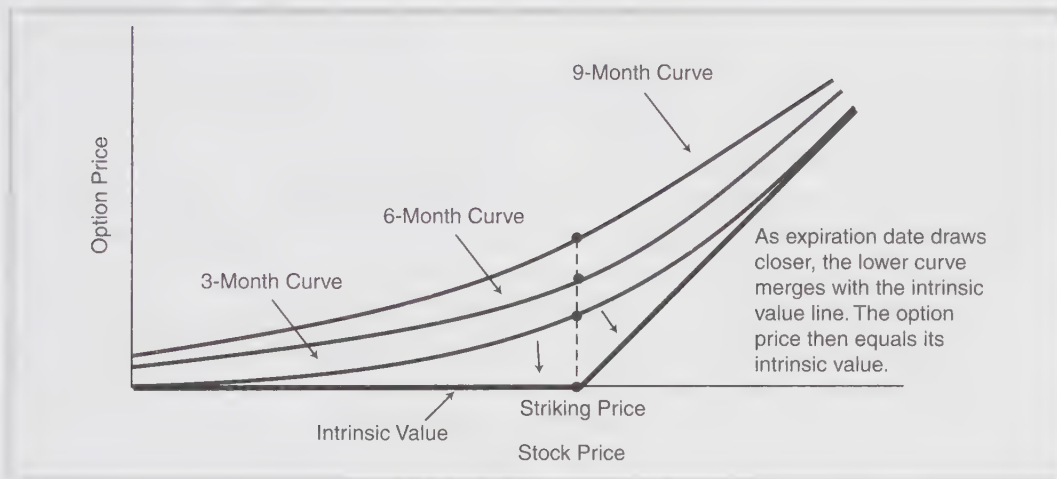
The certainty with which one expects the underlying stock to advance may also help to play a part in his selection of which call to buy. If one is fairly sure that the underlying stock is about to rise immediately, he should strive for more reward and not be as concerned about risk. This would mean buying short-term, slightly out-of-the-money calls. Of course, this is only a general rule; one would not normally buy an out-of-the-money call that has only one week remaining until expiration, in any case. At the opposite end of the spectrum, if one is very uncertain about his timing, he should buy the longest-term call, to moderate his risk in case his timing is wrong by a wide margin. This situation could easily result, for example, if one feels that a positive fundamental aspect concerning the company will assert itself and cause the stock to increase in price at an unknown time in the future. Since the buyer does not know whether this positive fundamental will come to light in the next month or six months from now, he should buy the longer-term call to allow room for error in timing.

In many cases, one is not intending to hold the purchased call for any significant period of time; he is just looking to capitalize on a quick, short-term movement by the underlying stock. In this case, he would want to buy a relatively short-term in-the-money call. Although such a call may be more expensive than an out-of-the-money call on the same underlying stock, it will most surely move up on any increase in price by the underlying stock. Thus, the short-term trader would profit.

THE DELTA

The reader should by now be familiar with basic facts concerning call options: The time premium is highest when the stock is at the striking price of the call; it is lowest deep in- or out-of-the-money; option prices do not decay at a linear rate—the time premium disappears more rapidly as the option approaches expiration. As a further means of review, the *option pricing curve* introduced in Chapter 1 is reprinted here. Notice that all the

FIGURE 3-1.
Option pricing curve; 3-, 6-, and 9-month calls.



facts listed above can be observed from Figure 3-1. The curves are much nearer the “intrinsic value” line at the ends than they are in the middle, implying that the time value premium is greatest when the stock is at the strike, and is least when the stock moves away from the strike either into- or out-of-the-money. Furthermore, the fact that the curve for the 3-month option lies only about halfway between the intrinsic value line and the curve of the 9-month option implies that the rate of decay of an at- or near-the-money option is not linear. The reader may also want to refer back to the graph of time value premium decay in Chapter 1 (Figure 1-4).

There is another property of call options that the buyer should be familiar with, the *delta* of the option (also called the *hedge ratio*). Simply stated, *the delta of an option is the amount by which the call will increase or decrease in price if the underlying stock moves by 1 point.*

Example: The delta of a call option is close to 1 when the underlying stock is well above the striking price of the call. If XYZ were 60 and the XYZ July 50 call were 10.10, the call would change in price by nearly 1 point if XYZ moved by 1 point, either up or down. A deeply out-of-the-money call has a delta of nearly zero. If XYZ were 40, the July 50 call might be selling at .25 of a point. The call would change very little in price if XYZ moved by one point, to either 41 or 39. When the stock is at the striking price, the delta is usually between .50 and .60. Thus, if XYZ were 50 and the XYZ July 50 call were 5, the call might increase to 5.50 if XYZ rose to 51 or decrease to 4.50 if XYZ dropped to 49. Very long-term calls may have even larger at-the-money deltas.

Actually, the delta changes each time the underlying stock changes even fractionally in price; it is an exact mathematical derivation that is presented in a later chapter. This is most easily seen by the fact that a deep in-the-money option has a delta of 1. However, if the stock should undergo a series of 1-point drops down to the striking price, the delta will be more like .50, certainly not 1 any longer. In reality, the delta changed instantaneously all during the price decline by the stock. For those who are geometrically inclined, the preceding option price curve is useful in determining a graphic representation of the delta. The delta is the slope of the tangent line to the price curve. Notice that a deeply in-the-money option lies to the upper right side of the curve, very nearly on the intrinsic value line, which has a slope of 1 above the strike. Similarly, a deeply out-of-the-money call lies to the left on the price curve, again near the intrinsic value line, which has a slope of zero below the strike.

Since it is more common to relate the option's price change to a full point change in the underlying stock (rather than to deal in "instantaneous" price changes), the concepts of *up delta* and *down delta* arise. That is, if the underlying stock moves up by 1 full point, a call with a delta of .50 might increase by .55. However, should the stock fall by one full point, the call might decrease by only .45. There is a different net price change in the call when the stock moves up by 1 full point as opposed to when it falls by a point. The up delta is observed to be .55 while the down delta is .45. In the true mathematical sense, there is only one delta and it measures "instantaneous" price change. The concepts of up delta and down delta are practical, rather than theoretical, concepts that merely illustrate the fact that the true delta changes whenever the stock price changes, even by as little as 1 point. In the following examples and in later chapters, only one delta is referred to.

The delta is an important piece of information for the call buyer because it can tell him how much of an increase or decrease he can expect for short-term moves by the underlying stock. This piece of information may help the buyer decide which call to buy.

Example: If XYZ is 47.50 and the call buyer expects a quick, but possibly limited, rise in price in the underlying stock, should he buy the 45 call or the 50 call? The delta may help him decide. He has the following information:

XYZ: 47.50	XYZ July 45 call:	price = 3.50,	delta = .65
	XYZ July 50 call:	price = 1.00,	delta = .25

It will make matters easier to make a slightly incorrect, but simplifying, assumption that the deltas remain constant over the short term. Which call is the better buy if the buyer expects the stock to quickly rise to 49? This would represent a 1.50-point increase in XYZ, which would translate into a 97-cent increase in the July 45 call price (1.50 times 0.65) or a 37-cent increase in the July 50 call (1.50 times 0.25). Consequently, the July 45 call, if it increased in price by 97 cents, would appreciate by 28%. The July 50 call, if it

increased by 37 cents, would appreciate by 37%. Thus, the July 50 appears to be the better buy in this simple example. Commissions should, of course, be included when making an analysis for actual investment.

The investor does not have to bother with computing deltas for himself. Any good call-buying data service will supply the information, and some brokerage houses provide this information free of charge.

More advanced applications of deltas are described in many of the succeeding chapters, as they apply to a variety of strategies.

WHICH OPTION TO BUY?

There are various trading strategies, some short-term, some long-term (even buy and hold). If one decides to use an option to implement a trading strategy, the *time horizon* of the strategy itself often dictates the general category of option that should be bought—in-the-money versus out-of-the-money, near-term versus long-term, etc. This statement is true whether one is referring to stock, index, or futures options. The general rule is this: *The shorter-term the strategy, the higher the delta should be of the instrument being used to trade the strategy.*

DAY TRADING

For example, day trading has become a popular endeavor. Statistics have been produced that indicate that most day traders lose money. In fact, there *are* profitable day traders; it simply requires more and harder work than many are willing to invest. Many day traders have attempted to use options in their strategies. These day traders apparently are attracted by the leverage available from options, but they often lose money via option trading as well.

What many of these option-oriented day traders fail to realize is that, for day-trading purposes, the instrument with the *highest* possible delta should be used. That instrument is the underlying, for it has a delta of 1.0. Day trading is hard enough without complicating it by trying to use options. So if you're day trading Microsoft (MSFT), trade the stock, not an option.

What makes options difficult in such a short-term situation is their relatively wide bid-asked spread, as compared to that of the underlying instrument itself. Also, a day trader is looking to capture only a small part of the underlying's daily move; an at-the-money or out-of-the-money option just won't respond well enough to those movements. That is, if the delta is too low, there just isn't enough room for the option day trader to make money.

If a day trader insists on using options, a short-term, in-the-money should be bought, for it has the largest delta available—preferably something approaching .90 or higher. This option will respond quickly to small movements by the underlying.

SHORT-TERM TRADING

Suppose one employs a strategy whereby he expects to hold the underlying for approximately a week or two. In this case, just as with day trading, a high delta is desirable. However, now that the holding period is more than a day, it may be appropriate to buy an option as opposed to merely trading the underlying, because the option lessens the risk of a surprisingly large downside move. Still, it is the short-term, in-the-money option that should be bought, for it has the largest delta, and will thus respond most closely to the movement in the underlying stock. Such an option has a very high delta, usually in excess of .80. Part of the reason that the high-delta options make sense in such situations is that one is fairly certain of the timing of day trading or very short-term trading systems. When the system being used for selection of which stock to trade has a high degree of timing accuracy, then the high-delta option is called for.

INTERMEDIATE-TERM TRADING

As the time horizon of one's trading strategy lengthens, it is appropriate to use an option with a lesser delta. This generally means that the timing of the selection process is less exact. One might be using a trading system based, for example, on sentiment, which is generally not an exact timing indicator, but rather one that indicates a general trend change at major turning points. The timing of the forthcoming move is not exact, because it often takes time for an extreme change in sentiment to reflect itself in a change of direction by the underlying.

Hence, for a strategy such as this, one would want to use an option with a smaller delta. The investor would limit his risk by using such an option, knowing that large moves are possible since the position is going to be held for several weeks or perhaps even a couple of months or more. Therefore, an at-the-money option can be used in such situations.

LONG-TERM TRADING

If one's strategy is even longer-term, an option with a lower delta can be considered. Such strategies would generally have only vague timing qualities, such as selecting a stock to buy based on the general fundamental outlook for the company. In the extreme, it would even apply to "buy and hold" strategies.

Generally, buying out-of-the-money options is not recommended; but for very long-term strategies, one might consider something *slightly* out-of-the-money, or at least a fairly long-term at-the-money option. In either case, that option will have a lower delta as compared to the options that have been recommended for the other strategies mentioned above. Alternatively, LEAPS options might be appropriate for stock strategies of this type.

ADVANCED SELECTION CRITERIA

The criteria presented previously represented elementary techniques for selecting which call to buy. In actual practice, one is not usually bullish on just one stock at a time. In fact, the investor would like to have a list of the “best” calls to buy at any given time. Then, using some method of stock selection, either technical or fundamental, he can select three or four calls that appear to offer the best rewards. This list should be ranked in order of the best potential rewards available, but the construction of the list itself is important.

Call option rankings for buying purposes must be based on the volatilities of the underlying stocks. This is not easy to do mathematically, and as a result many published rankings of calls are based strictly on percentage change in the underlying stock. Such a list is quite misleading and can lead one to the wrong conclusions.

Example: There are two stocks with listed calls: NVS, which is not volatile, and VVS, which is quite volatile. Since a call on the volatile stock will be higher-priced than a call on the nonvolatile stock, the following prices might exist:

NVS:	40	VVS:	40
NVS July 40 call:	2	VVS July 40 call:	4

If these two calls are ranked for buying purposes, based strictly on a percentage change in the underlying stock, the NVS call will appear to be the better buy. For example, one might see a list such as “best call buys if the underlying stock advances by 10%.” In this example, if each stock advanced 10% by expiration, both NVS and VVS would be at 44. Thus, the NVS July 40 would be worth 4, having doubled in price, for a 100% potential profit. Meanwhile, the VVS July 40 would be worth 4 also, for a 0% profit to the call buyer. This analysis would lead one to believe that the NVS July 40 is the better buy. Such a conclusion may be wrong, because an incorrect assumption was made in the ranking of the potentials of the two stocks. It is not right to assume that both stocks have the same probability of moving 10% by expiration. Certainly, the volatile stock has a much better

chance of advancing by 10% (or more) than the nonvolatile stock does. *Any ranking based on equal percentage changes in the underlying stock, without regard for their volatilities, is useless and should be avoided.*

The correct method of comparing these two July 40 calls is to utilize the actual volatilities of the underlying stocks. Suppose that it is known that the volatile stock, VVS, could expect to move 15% in the time to July expiration. The nonvolatile stock, NVS, however, could only expect a move of 5% in the same period. Using this information, the call buyer can arrive at the conclusion that VVS July 40 is the better call to buy:

Stock Price in July	Call Price
VVS: 46 (up 15%)	VVS July 40: 6 (up 50%)
NVS: 42 (up 5%)	NVS July 40: 2 (unchanged)

By assuming that each stock can rise in accordance with its volatility, we can see that the VVS July 40 has the better reward potential, despite the fact that it was twice as expensive to begin with. This method of analysis is much more realistic.

One more refinement needs to be made in this ranking process. Since most call purchases are made for holding periods of from 30 to 90 days, it is not correct to assume that the calls will be held to expiration. That is, even if one buys a 6-month call, he will normally liquidate it, to take profits or cut losses, in 1 to 3 months. *The call buyer's list should thus be based on how the call will perform if held for a realistic time period, such as 90 days.*

Suppose the volatile stock in our example, VVS, has the potential to rise by 12% in 90 days, while the less volatile stock, NVS, has the potential of rising only 4% in 90 days. In 90 days, the July 40 calls will not be at parity, because there will be some time remaining until July expiration. Thus, it is necessary to attempt to predict what their prices will be at the end of the 90-day holding period. Assume that the following prices are accurate estimates of what the July 40 calls will be selling for in 90 days, if the underlying stocks advance in relation to their volatilities:

Stock Price in 90 Days	Call Price
VVS: 44.8 (up 12%)	VVS July 40: 6 (up 50%)
NVS: 41.6 (up 4%)	NVS July 40: 2.50 (up 25%)

With some time remaining in the calls, they would both have time value premium at the end of 90 days. The bigger time premium would be in the VVS call, since the underlying stock is more volatile. Under this method of analysis, the VVS call is still the better one to buy.

The correct method of ranking potential reward situations for call buyers is as follows:

1. Assume each underlying stock can advance in accordance with its volatility over a fixed period (30, 60, or 90 days).
2. Estimate the call prices after the advance.
3. Rank all potential call purchases by highest percentage reward opportunity for aggressive purchases.
4. Assume each stock can decline in accordance with its volatility.
5. Estimate the call prices after the decline.
6. Rank all purchases by reward/risk ratio (the percentage gain from item 2 divided by the percentage loss from item 5).

The list from item 3 will generate more aggressive purchases because it incorporates potential rewards only. The list from item 6 would be a less speculative one. This method of analysis automatically incorporates the criteria set forth earlier, such as buying short-term out-of-the-money calls for aggressive purchases and buying longer-term in-the-money calls for a more conservative purchase. The delta is also a function of the volatility and is essentially incorporated by steps 1 and 4.

It is virtually impossible to perform this sort of analysis without a computer. The call buyer can generally obtain such a list from a brokerage firm or from a data service. For those individuals who have access to a computer and would like to generate such an analysis for themselves, the details of computing a stock's volatility and predicting the call prices are provided in Chapter 28 on mathematical techniques.

OVERPRICED OR UNDERPRICED CALLS

Formulae exist that are capable of predicting what a call should be selling for, based on the relationship of the stock price and the striking price, the time remaining to expiration, and the volatility of the underlying stock. These are useful, for example, in performing the second step in the foregoing analysis, estimating the call price after an advance in the underlying stock. In reality, a call's actual price may deviate somewhat from the price computed by the formula. If the call is actually selling for more than the "fair" (computed) price, the call is said to be overvalued. An *undervalued* call is one that is actually trading at a price that is less than the "fair" price.

If the calls are truly overpriced, there may be a strategy that can help reduce their cost while still preserving upside profit potential. This strategy, however, requires the

addition of a put spread to the call purchase, so it is beyond the scope of the subject matter at the current time. It is described in Chapter 23 on spreads combining calls and puts.

Generally, the amount by which a call is overvalued or undervalued may be only a small fraction of a point, such as 10 or 20 cents. In theory, the call buyer who purchases an undervalued call has gained a slight advantage in that the call should return to its “fair” value. However, in practice, this information is most useful only to market-makers or firm traders who pay little or no commissions for trading options. The general public cannot benefit directly from the knowledge that such a small discrepancy exists, because of commission costs.

One should not base his call buying decisions merely on the fact that a call is underpriced. It is small solace to the call buyer to find that he bought a “cheap” call that subsequently declined in price. The method of ranking calls for purchase that has been described does, in fact, give some slight benefit to underpriced calls. However, under the recommended method of analysis, a call will not automatically appear as an attractive purchase just because it is slightly undervalued.

TIME VALUE PREMIUM IS A MISNOMER

This is a topic that will be mentioned several times throughout the book, most notably in conjunction with volatility trading. It is introduced here because even the inexperienced option trader must understand that the portion of an option’s price that is *not* intrinsic value—the part that we routinely call “time value premium”—is really composed of much more than just time value. Yes, time will eventually wear away that portion of the option’s price as expiration approaches. However, when an option has a considerable amount of time remaining until its expiration, the more important component of the option value is really volatility. If traders expect the underlying stock to be volatile, the option will be expensive; if they expect the opposite, the option will be cheap. This expensiveness and cheapness is reflected in the portion of the option that is not intrinsic value. For example, a six-month option will not decay much in one day’s time, but a quick change in volatility expectations by option traders can heavily affect the price of the option, especially one with a good deal of time remaining. So an option buyer should carefully assess his purchases, not just view them as something that will waste away. With careful analysis, option buyers can do very well, if they consider what can happen *during* the life of the option, and not merely what will happen at expiration.

CALL BUYERS’ FRUSTRATIONS

Despite one’s best efforts, it may often seem that one does not make much money when a fairly volatile stock makes a quick move of 3 or 4 points. The reasons for this are somewhat

more complex than can be addressed at this time, although they relate strongly to delta, time decay, and the volatility of the underlying stock. They are discussed in Chapter 36, “The Basics of Volatility Trading.” If one plans to conduct a serious call buying strategy, he should read that chapter before embarking on a program of extensive call buying.

FOLLOW-UP ACTION

The simplest follow-up action that the call buyer can implement when the underlying stock drops is to sell his call and cut his losses. There is often a natural tendency to hold out hope that the stock can rally back to or above the striking price. Most of the time, the buyer does best by cutting his losses in situations in which the stock is performing poorly. He might use a “mental” stop price or could actually place a sell stop order, depending on the rules of the exchange where the call is traded. In general, stop orders for options result in poor executions, so using a “mental” stop is better. That is, one should base his exit point on the technical pattern of the underlying stock itself. If it should break down below support, for example, then the option holder should place a market (not held) order to sell his call option.

If the stock should rise, the buyer should be willing to take profits as well. Most buyers will quite readily take a profit if, for example, a call that was bought for 5 points had advanced to be worth 10 points. However, the same investor is often reluctant to sell a call at 2 that he had previously bought for 1 point, because “I’ve only made a point.” The similarity is clear—both cases resulted in approximately a 100% profit—and the investor should be as willing to accept the one as he is the other. This is not to imply that all calls that are bought at 1 should be sold when and if they get to 2, but the same factors that induce one to sell the 10-point call after doubling his money should apply to the 2-point call as well.

In fact, taking partial profits after a call holding has increased in value is often a wise plan. For example, if someone bought a number of calls at a price of 3, and they later were worth 5, it might behoove the call holder to sell one-third to one-half of his position at 5, thereby taking a partial profit. Having done that, it is often easier to let the profits run on the balance, and letting profits run is generally one of the keys to successful trading.

It is rarely to the call buyer’s benefit to exercise the call if he has to pay commissions. When one exercises a call, he pays a stock commission to buy the stock at the striking price. Then when the stock is sold, a stock sale commission must also be paid. Since option commissions are much smaller, dollarwise, than stock commissions, the call holder will usually realize more net dollars by selling the call in the option market than by exercising it.

LOCKING IN PROFITS

When the call buyer is fortunate enough to see the underlying stock advance relatively quickly, he can implement a number of strategies to enhance his position. These strategies are often useful to the call buyer who has an unrealized profit but is torn between taking the profit or holding on in an attempt to generate more profits if the underlying stock should continue to rise.

Example: A call buyer bought an XYZ October 50 call for 3 points when the stock was at 48. Then the stock rises to 58. The buyer might consider selling his October 50 (which would probably be worth about 9 points) or possibly taking one of several actions, some of which might involve the October 60 call, which may be selling for 3 points. Table 3-1 summarizes the situation. At this point, the call buyer might take one of four basic actions:

1. Liquidate the position by selling the long call for a profit.
2. Sell the October 50 that he is currently long and use part of the proceeds to purchase October 60's.
3. Create a spread by selling the October 60 call against his long October 50.
4. Do nothing and remain long the October 50 call.

TABLE 3-1.
Present situation on XYZ October calls.

Original Trade	Current Prices
XYZ common: 48	XYZ Common: 58
Bought XYZ October 50 at 3	XYZ October 50: 9
	XYZ October 60: 3

Each of these actions would produce different levels of risk and reward from this point forward. If the holder sells the October 50 call, he makes a 6-point profit, less commissions, and terminates the position. He can realize no further appreciation from the call, nor can he lose any of his current profits; he has realized a 6-point gain. *This is the least aggressive tactic of the four.* If the underlying stock continues to advance and rises above 63, any of the other three strategies will outperform the complete liquidation of the call. However, if the underlying stock should instead decline below 50 by expiration, this action would have provided the most profit of the four strategies.

The other simple tactic, the fourth one listed, is to do nothing. If the call is then held

to expiration, *this tactic would be the riskiest of the four*. It is the only one that could produce a loss at expiration if XYZ fell back below 50. However, if the underlying stock continues to rise in price, more profits would accrue on the call. Every call buyer realizes the ramifications of these two tactics—liquidating or doing nothing—and is generally looking for an alternative that might allow him to reduce some of his risk without cutting off his profit potential completely. The remaining two tactics are geared to this purpose: limiting the total risk while providing the opportunity for further profits of an amount greater than those that could be realized by liquidating.

The strategy in which the holder sells the call that he is currently holding, the October 50, and uses part of the proceeds to buy the call at the next higher strike is called *rolling up*. In this example, he could sell the October 50 at 9, pocket his initial 3-point investment, and use the remaining proceeds to buy two October 60 calls at 3 points each. Thus, it is sometimes possible for the speculator to recoup his entire original investment and still increase the number of calls outstanding by rolling up. Once this has been done, the October 60 calls will represent pure profits, whatever their price. *The buyer who “rolls up” in this manner is essentially speculating with someone else’s money*. He has put his own money back in his pocket and is using accrued profits to attempt to realize further gains. At expiration, this tactic would perform best if XYZ increased by a substantial amount. This tactic turns out to be the worst of the four at expiration if XYZ remains near its current price, staying above 53 but not rising above 63 in this example.

The other alternative, the third one listed, is to continue to hold the October 50 call but to sell the October 60 call against it. This would create what is known as a bull spread, and the tactic can be used only by traders who have a margin account and can meet their firm’s minimum equity requirement for spreading (generally \$2,000). This spread position has no risk, for the long side of the spread—the October 50—cost 3 points, and the short side of the spread—the October 60—brought in 3 points via its sale. Even if the underlying stock drops below 50 by expiration and all the calls expire worthless, the trader cannot lose anything except commissions. On the other hand, the maximum potential of this spread is 10 points, the difference between the striking prices of 50 and 60. This maximum potential would be realized if XYZ were anywhere above 60 at expiration, for at that time the October 50 call would be worth 10 points more than the October 60 call, regardless of how far above 60 the underlying stock had risen. *This strategy will be the best performer of the four if XYZ remains relatively unchanged*, above the lower strike but not much above the higher strike by expiration. It is interesting to note that this tactic is *never the worst performer of the four tactics*, no matter where the stock is at expiration. For example, if XYZ drops below 50, this strategy has no risk and is therefore better than the “do nothing” strategy. If XYZ rises substantially, this spread produces a profit of 10 points, which is better than the 6 points of profit offered by the “liquidate” strategy.

* * *

There is no definite answer as to which of the four tactics is the best one to apply in a given situation. However, if a call can be sold against the currently long call to produce a bull spread that has little or no risk, it may often be an attractive thing to do. It can never turn out to be the worst decision, and it would produce the largest profits if XYZ does not rise substantially or fall substantially from its current levels. Tables 3-2 and 3-3 summarize the four alternative tactics, when a call holder has an unrealized profit. The four tactics, again, are:

TABLE 3-2.
Comparison of the four alternative strategies.

If the underlying stock then . . .	The best tactic was . . .	And the worst tactic was . . .
continues to rise dramatically . . .	"roll up"	liquidate
rises moderately above the next strike . . .	do nothing	liquidate or "roll up"
remains relatively unchanged . . .	spread	"roll up"
falls back below the original strike . . .	liquidate	do nothing

TABLE 3-3.
Results at expiration.

XYZ Price at Expiration	"Roll-up" Profit	"Do Nothing" Profit	"Spread" Profit	Liquidating Profit
50 or below	\$ 0	-\$ 300(W)	\$ 0	+\$600 (B)
53	0(W)	0(W)	+ 300	+ 600 (B)
56	0(W)	+ 300	+ 600(B)	+ 600 (B)
60	0(W)	+ 700	+ 1,000(B)	+ 600
63	+ 600(W)	+ 1,000 (B)	+ 1,000(B)	+ 600(W)
67	+ 1,400 (B)	+ 1,400 (B)	+ 1,000	+ 600(W)
70	+ 2,000 (B)	+ 1,700	+ 1,000	+ 600(W)

1. "Do nothing"—continue to hold the currently long call.
2. "Liquidate"—sell the long call to take profits and do not reinvest.
3. "Roll up"—sell the long call, pocket the original investment, and use the remaining proceeds to purchase as many out-of-the-money calls as possible.

4. “Spread”—create a bull spread by selling the out-of-the-money call against the currently profitable long call, preferably taking in at least the original cost of the long call.

Note that each of the four tactics proves to be the best tactic in one case or another, but that the spread tactic is never the worst one. Tables 3-2 and 3-3 represent the results from holding until expiration. For those who prefer to see the actual numbers involved in making these comparisons between the four tactics, Table 3-3 summarizes the potential profits and losses of each of the four tactics using the prices from the example above. “W” indicates that the tactic is the worst one at that price, and “B” indicates that it is the best one.

There are, of course, modifications that an investor might make to any of these tactics. For example, he might decide to sell out half of his long call position, recovering a major part of his original cost, and continue to hold the remainder of the long calls. This still leaves room for further appreciation.

Selling a portion of the position and retaining the rest is known as “taking a partial profit.” This tactic and all of the above ones—except “do nothing”—can be harmful to a position in which the underlying moves strongly in one’s favor for an extended distance. Every time one takes partial profits, rolls up, or takes other measures, he is doing something bearish to his position. Those little bearish actions will be harmful if the underlying continues to perform favorably. There is a school of thought that says one is better off merely using a trailing stop—a stop placed at the 20-day moving average of the underlying, say—than to make bearish adjustments to a bullish position. Of course, in a choppy market that is swinging up and down, partial profits may prove to be preferable, but a call buyer’s best way to maximize profits over time is to be lucky or good enough to be long in a strong trend. He should do everything he can to maximize the profits and ride that trend, which would be “do nothing” and use a stop that trails along behind the market.

DEFENSIVE ACTION

Two follow-up strategies are sometimes employed by the call buyer when the underlying stock declines in price. Both involve spread strategies; that is, being long and short two different calls on the same underlying stock simultaneously. Spreads are discussed in detail in later chapters. This discussion of spreads applies only to their use by the call buyer.

“Rolling Down.” If an option holder owns an option at a currently unrealized loss, it may be possible to greatly increase the chances of making a limited profit on a relatively small rebound in the stock price. In certain cases, the investor may be able to implement such a strategy at little or no increase in risk.

Many call buyers have encountered a situation such as this: An XYZ October 35 call was originally bought for 3 points in hopes of a quick rise in the stock price. However, because of downward movements in the stock—to 32, say—the call is now at 1.50 with October expiration nearer. If the call buyer still expects a mild rally in the stock before expiration, he might either hold the call or possibly “average down” (buy more calls at 1.50). In either case he will need a rally to nearly 38 by expiration in order to break even. Since this would necessitate at least a 15% upward move by the stock before expiration, it cannot be considered very likely. Instead, the buyer should consider implementing the following strategy, which will be explained through the use of an example.

Example: The investor is long the October 35 call at this time:

XYZ, 32;
XYZ October 35 call, 1.50; and
XYZ October 30 call, 3.

One could sell two October 35's and, at the same time, buy one October 30 for no additional investment before commissions. That is, the sale of 2 October 35's at \$150 each would bring in \$300, exactly the cost, before commissions, of buying the October 30 call. *This is the key to implementing the roll-down strategy: that one be able to buy the lower strike call and sell two of the higher strike calls for nearly even money.*

Note that the investor is now short the call that he previously owned, the October 35. Where he previously owned one October 35, he has now sold two of them. He is also now long one October 30 call. Thus, his position is:

long 1 XYZ October 30 call,
short 1 XYZ October 35 call.

This is technically known as a *bull spread*, but the terminology is not important. Table 3-4 summarizes the transactions that the buyer has made to acquire this spread. The trader now “owns” the spread at a cost of \$300, plus commissions. By making this trade, he has lowered his break-even point significantly without increasing his risk. However, the maximum profit potential has also been limited; he can no longer capitalize on a strong rebound by the underlying stock.

In order to see that the break-even point has been lowered, consider what the results are if XYZ is at 33 at October expiration. The October 30 call would be worth 3 points and the October 35 would expire worthless with XYZ at 33. Thus, the October 30 call could be sold to bring in \$300 at that time, and there would not be any expense to buy

TABLE 3-4.
Transactions in bull spread.

	Trade	Cost before Commissions
Original trade	Buy 1 October 35 call at 3	\$300 debit
Later trade	Sell 2 October 35 calls at 1.50	\$300 credit
	Buy 1 October 30 call at 3	\$300 debit
Net position	Long 1 October 30 call	\$300 debit
	Short 1 October 35 call	

back the October 35. Consequently, the spread could be liquidated for \$300, exactly the amount for which it was “bought.” The spread then breaks even at 33 at expiration. If the call buyer had not rolled down, his break-even point would be 38 at expiration, for he paid 3 points for the original October 35 call and he would thus need XYZ to be at 38 in order to be able to liquidate the call for 3 points. Clearly, the stock has a better chance of recovering to 33 than to 38. *Thus, the call buyer significantly lowers his break-even point by utilizing this strategy.*

Lowering the break-even point is not the investor’s only concern. He must also be aware of what has happened to his profit and loss opportunities. The risk remains essentially the same—the \$300 in debits, plus commissions, that has been paid out. The risk has actually increased slightly, by the amount of the commissions spent in “rolling down.” However, the stock price at which this maximum loss would be realized has been lowered. With the original long call, the October 35, the buyer would lose the entire \$300 investment anywhere below 35 at October expiration. The spread strategy, however, would result in a total loss of \$300 only if XYZ were below 30 at October expiration. With XYZ above 30 in October, the long side of the spread could be liquidated for some value, thereby avoiding a total loss. *The investor has reduced the chance of realizing the maximum loss, since the stock price at which that loss would occur has been lowered by 5 points.*

As with most investments, the improvement of risk exposure—lowering the break-even point and lowering the maximum loss price—necessitates that some potential reward be sacrificed. In the original long call position (the October 35), the maximum profit potential was unlimited. In the new position, the potential profit is limited to 2 points if XYZ should rally back to, *or anywhere above*, 35 by October expiration. To see this, assume XYZ is 35 at expiration. Then the long October 30 call would be worth 5 points, while the October 35 would expire worthless. Thus, the spread could be liquidated for 5

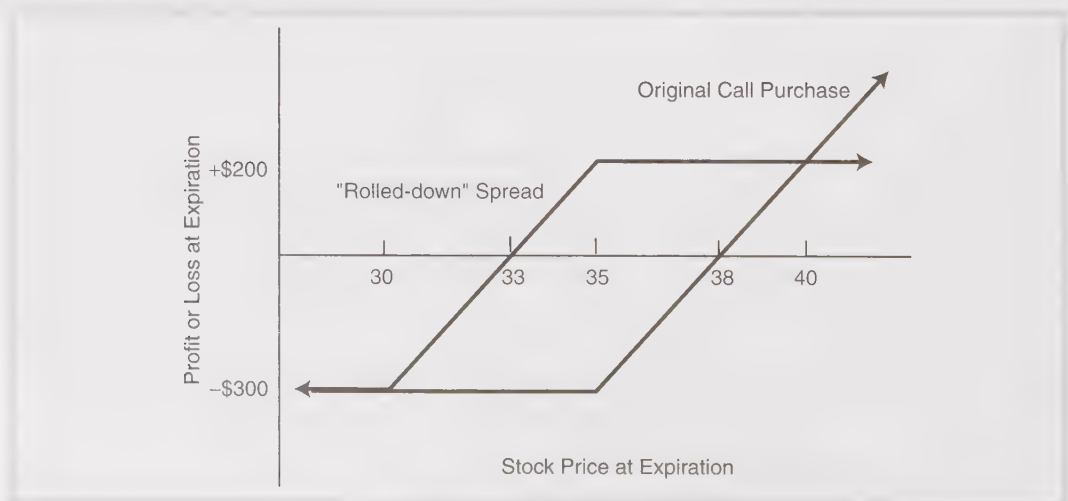
points, a 2-point profit over the 3 points paid for the spread. This is the limit of profit for the spread, however, since if XYZ is above 35 at expiration, any further profits in the long October 30 call would be offset by a corresponding loss on the short October 35 call. Thus, if XYZ were to rally heavily by expiration, the “rolled down” position would not realize as large a profit as the original long call position would have realized.

Table 3-5 and Figure 3-2 summarize the original and new positions. Note that the new position is better for stock prices between 30 and 40. Below 30, the two positions are

TABLE 3-5.
Original and spread positions compared.

Stock Price at Expiration	Long Call Result	Spread Result
25	-\$300	-\$300
30	- 300	- 300
33	- 300	0
35	- 300	+ 200
38	0	+ 200
40	+ 200	+ 200
45	+ 700	+ 200

FIGURE 3-2.
Companion: original call purchase vs. spread.



equal, except for the additional commissions spent. If the stock should rally back above 40, the original position would have worked out better. *The new position is an improvement, provided that XYZ does not rally back above 40 by expiration.* The chances that XYZ could rally 8 points, or 25%, from 32 to 40 would have to be considered relatively remote. Rolling the long call down into the spread would thus appear to be the correct thing to do in this case.

This example is particularly attractive, because no additional money was required to establish the spread. In many cases, however, one may find that the long call cannot be rolled into the spread at even money. Some debit may be required. This fact should not necessarily preclude making the change, since a small additional investment may still significantly increase the chance of breaking even or making a profit on a rebound.

Example: The following prices now exist, rather than the ones used earlier. Only the October 30 call price has been altered:

XYZ, 32;

XYZ October 35 call, 1.50; and

XYZ October 30 call, 4.

With these prices, a 1-point debit would be required to roll down. That is, selling 2 October 35 calls would bring in \$300 (\$150 each), but the cost of buying the October 30 call is \$400. Thus, the transaction would have to be done at a cost of \$100, plus commissions. With these prices, the break-even point after rolling down would be 34, still well below the original break-even price of 38. The risk has now been increased by the additional 1 point spent to roll down. If XYZ should drop below 30 at October expiration, the investor would have a total loss of 4 points plus commissions. The maximum loss with the original long October 35 call was limited to 3 points plus a smaller amount of commissions. Finally, the maximum amount of money that the spread could make is now \$100, less commissions. The alternative in this example is not nearly as attractive as the previous one, but it might still be worthwhile for the call buyer to invoke such a spread if he feels that XYZ has limited rally potential up to October expiration.

One should not automatically discard the use of this strategy merely because a debit is required to convert the long call to a spread. Note that to “average down” by buying an additional October 35 call at 1.50 would require an additional investment of \$150. This is more than the \$100 required to convert into the spread position in the immediately preceding example. The break-even point on the position that was “averaged down” would be over 37 at expiration, whereas the break-even point on the spread is 34. Admittedly, the averaged-down

position has much more profit potential than the spread does, but the conversion to the spread is less expensive than “averaging down” and also provides a lower break-even price.

In summary, then, if the call buyer finds himself with an unrealized loss because the stock has declined, and yet is unwilling to sell, he may be able to improve his chances of breaking even by “rolling down” into a spread. That is, he would sell 2 of the calls that he is currently long—the one that he owns plus another one—and simultaneously buy one call at the next lower striking price. If this transaction of selling 2 calls and buying 1 call can be done for approximately even money, it could definitely be to the buyer’s benefit to implement this strategy, because the break-even point would be lowered considerably and the buyer would have a much better chance of getting out even or making a small profit should the underlying stock have a small rebound.

Creating a Calendar Spread. A different type of defensive spread strategy is sometimes used by the call buyer who finds that the underlying stock has declined. In this strategy, the holder of an intermediate- or long-term call sells a near-term call, with the same striking price as the call he already owns. This creates what is known as a calendar spread. The idea behind doing this is that if the short-term call expires worthless, the overall cost of the long call will be reduced to the buyer. Then, if the stock should rally, the call buyer has a better chance of making a profit.

Example: Suppose that an investor bought an XYZ October 35 call for 3 points sometime in April. By June the stock has fallen to 32, and it appears that the stock might remain depressed for a while longer. The holder of the October 35 call might consider selling a July 35 call, perhaps for a price of 1 point. Should XYZ remain below 35 until *July* expiration, the short call would expire worthless, earning a small, 1-point profit. The investor would still own the October 35 call and would then hope for a rally by XYZ before October in order to make profits on that call. Even if XYZ does not rally by October, he has decreased his overall loss by the amount received for the sale of the July 35 call.

This strategy is not as attractive to use as the previous one. If XYZ should rally before July expiration, the investor might find himself with two losing positions. For example, suppose that XYZ rallied back to 36 in the next week. His short call that he sold for 1 point would be selling for something more than that, so he would have an unrealized loss on the short July 35. In addition, the October 35 would probably not have appreciated back to its original price of 3, and he would therefore have an unrealized loss on that side of the spread as well.

Consequently, this strategy should be used with great caution, for if the underlying stock rallies quickly before the near-term expiration, the spread could be at a loss on both

sides. Note that in the former spread strategy, this could not happen. Even if XYZ rallied quickly, some profit would be made on the rebound.

A FURTHER COMMENT ON SPREADS

Anyone not familiar with the margin requirements for spreads, under both the exchange margin rules and the rules of the brokerage firm he is dealing with, should not attempt to utilize a spread transaction. Later chapters on spreads outline the more common requirements for spread transactions. In general, one must have a margin account to establish a spread and must have a minimum amount of equity in the account. Thus, the call buyer who operates in a cash account cannot necessarily use these spread strategies. To do so might incur a margin call and possible restriction of one's trading account. Therefore, check on specific requirements before utilizing a spread strategy. Do not assume that a long call can automatically be "rolled" into any sort of spread.